

December 2025 CSCA

Select this question (there are 4 options per question, and only 1 is correct)

- 1 Which of the following physical quantities is not a vector?
- A. intensity of magnetization B. momentum
C. electric field intensity D. potential energy
2. The gravitational force acting on an object on Mars is one-third of its gravitational force on Earth. Therefore, the gravitational force on an object with mass $m=4.8\text{kg}$ on Mars is (). ($g=10\text{ m/s}^2$)
- A. 144N B. 24N
C. 48N D. 16N
3. A spring with a spring constant $k=100\text{ N/m}$ is stretched from its original length of 10cm to 15cm under a tension force F . What is the magnitude of F ?
- A.5N B.25N
C.10N D.15N
4. An object with mass $m=4\text{kg}$ is stationary on a horizontal surface, where the sliding friction coefficient between the surface and the object is $\mu=0.1$. When a horizontal force of 2N is applied to the object, the frictional force exerted on it is ().
- A. 2N B.3N
C.4N D.6N
5. An object is subjected to three forces with magnitudes $1 = 2$, $2 = 4$, and $3 = 6$. The maximum resultant force acting on the object is ().
- A.12N B.0N
C.10N D.24N
- 6 On a smooth horizontal surface, an object with mass $m=2\text{kg}$ is subjected to a horizontal force F , starting from rest and undergoing uniform acceleration. If its displacement within 2 seconds is 8m, then the magnitude of the horizontal force F is ().
- A.10N B. 6N
C.8N D.2N
- 7 An object moving in uniform acceleration on a smooth horizontal surface from rest has an acceleration of $a=2\text{m/s}^2$. What is its velocity at the end of 4 seconds?
- A. 4m/s B.16m/s
C. 8m/s D.2m/s
- 8 An object is dropped from a height of 100 meters. What is its height from the ground at 4 seconds? ($g=10\text{ m/s}^2$)
- A.45m B.10m
C. 80m D.20m
- 9 An object with mass $m=5\text{kg}$ is placed on a long smooth inclined plane with an inclination angle $\theta=30^\circ$. Starting from rest, the distance it travels in 2 seconds is (). ($g=10\text{m/s}^2$)
- A.5m B.1m
C.10m D.2m
10. For a 2kg object subjected to two forces of 6N and 8N respectively, what is its maximum acceleration?
- A.4m/s² B.7m/s²
C.3m/s² D.5m/s²
11. A 0.5-meter-long straight conductor is placed perpendicular to a uniform magnetic field. When a 10A current flows through it, the conductor experiences an Ampere force of 1.5N. Determine the magnetic induction intensity of the field.
- A.0.3T B.0.2T
C.0.1T D.0.4T

12. A point moves in uniform circular motion with a velocity of $v=6\text{ m/s}$ and a radius of $R=3\text{ m}$. What is its angular velocity?

A. $2\pi\text{rad/s}$

B. $3\pi\text{rad/s}$

C. 2rad/s

D. 3rad/s

13 As shown in the figure, an object moves from point A to point C via point B along two semicircles with a radius of $R=6\text{ m}$ in a time of $t=4\text{ s}$. What is the average velocity of the object?



A. 6 m/s

B. 12 m/s

C. $6\pi\text{ m/s}$

D. $3\pi\text{ m/s}$

14. Which of the following statements about distance and displacement is correct?

A. Distance is a vector, displacement is a scalar.

B. The magnitude of displacement is not necessarily equal to the distance traveled.

C. Both distance and displacement are vectors

D. The magnitude of displacement can be equal to the distance traveled.

15. An object with mass $m=0.2\text{ kg}$ moves at constant velocity $v=20\text{ m/s}$ on a smooth horizontal surface. What is its momentum?

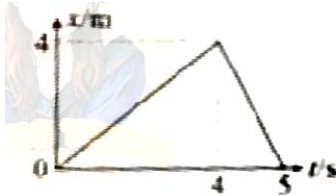
A. $0.8\text{ kg} \cdot \text{m/s}$

B. $4\text{ kg} \cdot \text{m/s}$

C. $0.4\text{ kg} \cdot \text{m/s}$

D. $8\text{ kg} \cdot \text{m/s}$

16. The graph of the displacement (x) of a toy car moving in a straight line versus time (t) is shown in the figure. Which of the following statements is correct?



A. Within 0 to 4 seconds, the toy car moves a distance of 8 meters. B. Between 4 and 5 seconds, the toy car moves at a speed of 0.8 meters per second.

C. The toy car's speed is 1.0 m/s from 0 to 4 seconds. D. The toy car's average speed is 0.8 m/s from 0 to 5 seconds.

17. An object with mass $m=5\text{ kg}$ is placed on a smooth inclined plane with an angle of $\theta=30^\circ$ and a height of 20 m. Starting from rest, the gravitational impulse it experiences in 2 seconds is (). ($g=10\text{ m/s}^2$)

A. $50\text{ N} \cdot \text{s}$

B. $25\text{ N} \cdot \text{s}$

C. $200\text{ N} \cdot \text{s}$

D. $100\text{ N} \cdot \text{s}$

18. When a 10kg object is pulled horizontally by a force F of 20N, it moves 5m on a rough horizontal surface with a frictional force $f=5\text{ N}$. The work done by the pulling force F is ().

A. 100J

B. 75J

C. -25J

D. 25J

19 A 3kg ball on a smooth surface is subjected to an external force that increases its velocity from 2 m/s to 6 m/s. Calculate the work done by this force.

A. 54J

B. 60J

C. 48J

D. 6J

2. When light enters a medium from air at an incident angle $i=60^\circ$, the refractive angle $r=30^\circ$, and the refractive index of air is $n=1$, the refractive index of the medium is ().

- A. 2 B. 3
C. 5 D. 1.5

21. A spring oscillator undergoing simple harmonic motion completes one full cycle in 2 seconds. What is its frequency?

- A. 2 Hz B. 0.5 Hz
C. 0.2 Hz D. 5 Hz

22. A sound wave with a frequency of $f=200$ Hz travels through air at a speed of $v=340$ m/s. What is the wavelength of this sound wave?

- A. 1.7m B. 3.4m
C. 6.8m D. 680m

23. Each question has 4 options, and only 1 is correct.

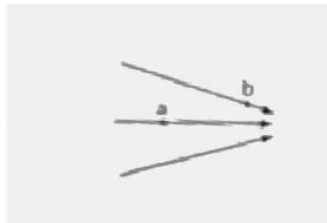
In a vacuum, two point charges are separated by a distance d , with a Coulomb force of F . If the distance between the charges is kept constant and the charge of one of them is doubled, the Coulomb force between them becomes ().

- (A) 2 times
(B) 4 times
A. 2 times B. 4 times
C. $\frac{1}{4}$ times D. $\frac{1}{2}$ times

24. The velocity of a water drop with zero initial velocity falling from a height of 5m is ($g = 10$ m/s²) ()

- A. 10m/s B. 5m/s
C. 2m/s D. 20m/s

25. The electric field lines are distributed as shown in the figure. If equal point charges are placed at points a and b within the electric field, the relationship between the electric forces acting on these charges is ().



- A. $\geq$ B. =
C. < D. >

The total resistance of two resistors, 2620Ω and 30Ω , in parallel is ().

- A. 50Ω B. 60Ω
C. 10Ω D. 12Ω

27 Two resistors with resistance values of $R_1=40\Omega$ and $R_2=30\Omega$ are connected in parallel and connected to a circuit with a voltage of 12V. The current ratio $I_1:I_2$ through the two resistors R_1 and R_2 is ()

- A. 1:1 B. 1:2
C. 4:3 D. 3:4

28. For an ideal gas of a given mass, if the temperature remains constant and the pressure decreases, its volume will ().

- A. invariant B. diminution
C. All of the above answers are possible. D. largen

29. A point charge of $3.0 \times 10^{-9} \text{ C}$ is placed in an electric field of $6.0 \times 10^4 \text{ N/C}$. The magnitude of the electric force acting on the point charge is ().

- A. $1.8 \times 10^{-4} \text{ N}$ B. $2.0 \times 10^{-4} \text{ N}$
C. $1.8 \times 10^4 \text{ N}$ D. $1.8 \times 10^{-5} \text{ N}$

30. The international units of magnetic induction B and magnetic flux are ().

- A. Tesla, Weber B. Amperes, volts
C. Pascal, Hertz D. Joules, watts

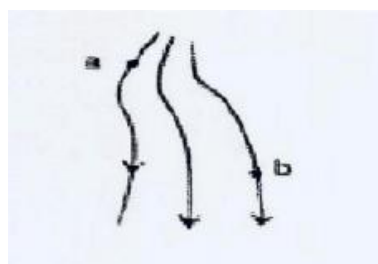
31. When a beam of light strikes the air-glass interface at an angle of 30° , the angle between the reflected and incident rays is ().

- A. 30° B. 45°
C. 60° D. 80°

32. A 0.2-meter-long wire carrying a constant current is placed in a uniform magnetic field with a magnetic induction intensity of 0.8 T. If the maximum ampere force acting on the wire in the magnetic field is 0.08 N, the current flowing through the wire is ().

- A. 0.5A B. 5A
C. 0.2A D. 2A

33. An ideal gas of a certain mass changes from state A to state B. The process $A \rightarrow B$ is an isobaric process (constant pressure). Given that the volume of state A is $V_1 = 1.5 \text{ m}^3$, the temperature of state A is $T_1 = 300 \text{ K}$, and the temperature of state B is $T_2 = 400 \text{ K}$, the volume of state B is ()



- A. 1.5 m^3 B. 4.5 m^3
C. 2 m^3 D. 6 m^3

35. A ball of mass $m_1 = 20 \text{ kg}$ is moving at $v = 6 \text{ m/s}$ and collides with a car that is stationary on a smooth horizontal surface. The car has a mass of 40 kg. What is the combined velocity of the ball and the car after the collision?

- A. 2 m/s B. 3 m/s
C. 4 m/s D. 5 m/s

36. A basketball with a mass of 0.5 kg is launched from the ground to a height of 1 meter, then falls back to the ground. The change in gravitational potential energy during this entire process is ().

- A. 5J B. 0.5J
C. 10J D. 0J

37. As shown in the figure, a thin rope is fixed at point O at one end and suspended from the other end a ball A with a mass of $m = 6 \text{ kg}$. When the rope forms a 30-degree angle with the vertical direction and the ball A is at rest, the horizontal force applied to the ball A should be () ($g = 10 \text{ m/s}^2$).

- A. 10 N B. 12 N
C. 20 N D. 20 N

38. The diagram shows two charged metal spheres A and B in a vacuum, each carrying a charge of $Q = \pm 3.0 \times 10^{-4} \text{ C}$, with a distance of $r = 6 \text{ cm}$ between them. Given the electrostatic constant $k = 9 \times 10^9 \text{ N} \cdot \text{m}^2 / \text{C}^2$, the electric field strength at the midpoint between the spheres is

- A. $1.5 \times 10^4 \text{ N/C}$ B. 0 N/C
C. $6.0 \times 10^4 \text{ N/C}$ D. $3.0 \times 10^4 \text{ N/C}$

39. A small ball is thrown horizontally from a building 20m high at a speed of $v_0 = 20\text{m/s}$. The air resistance is not considered. Find the velocity of the ball when it lands. ($g = 10\text{m/s}^2$)

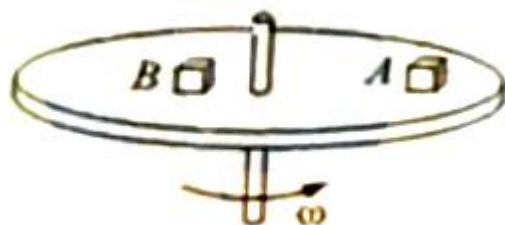
A. $40\sqrt{2}\text{m/s}$

B. $20\sqrt{2}\text{m/s}$

C. 40m/s

D. 20m/s

40. As shown in the figure, two objects A and B with equal mass are placed on a horizontal disk rotating uniformly around a vertical axis. Since both objects remain stationary relative to the disk, the two objects ().



A. The angular velocity is the same.

The angular velocity of B is greater than that of A.

C. The angular velocity of A is less than that of B. D. The linear velocity is the same.

41. A point charge with a charge of $+4 \times 10^{-6}\text{C}$ is moved from point A to point B in an electric field. The work done by the electric field force is

$-8 \times 10^{-4}\text{J}$, then the potential difference between AB is ()

A. -200V

B. 200V

C. -2000V

D. 2000V

42. Consider a point charge conductor A carrying $8q$ positive charge. At a distance r from A, there is another identical conductor B carrying $-4q$ negative charge. When the two conductors are brought into contact and then returned to their original positions, the Coulomb force between them will be restored to its original value ().

B. 4 times

A. $\frac{1}{4}$ times

C. 8 times

D. $\frac{1}{8}$ times

43. A bullet traveling at 700m/s horizontally decelerates to 500m/s after penetrating the first fixed wooden board. If it continues through a second identical wooden board, its velocity becomes ().

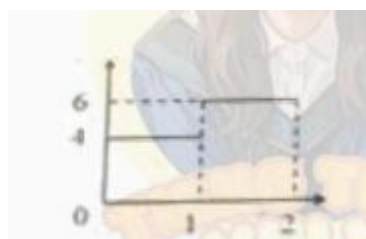
A. 200m/s

B. 100m/s

C. 50m/s

D. 300m/s

44. A 2kg object starts from rest on a smooth horizontal surface and is subjected to a horizontal force. The graph of the force versus time is shown in the figure. What is the object's kinetic energy at the end of 2 seconds? (A) 25 (B) 16 (C) 36 (D) 50



A. 25

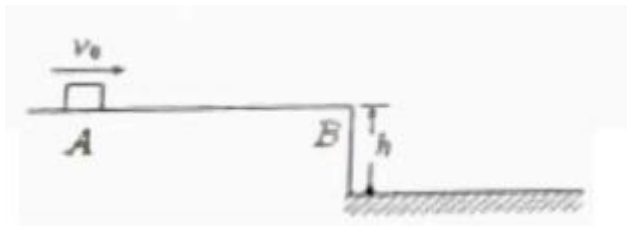
B. 16

C. 36

D. 50

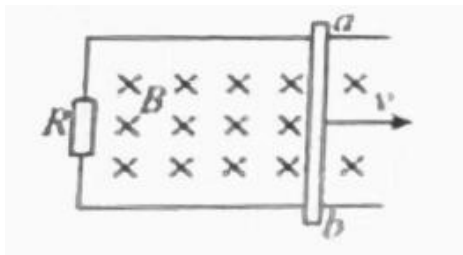
45. As shown in the figure, the horizontal platform AB is $h = 0.80\text{m}$ above the ground. A slider moves in uniform acceleration along the platform from point A with an initial velocity of $v_0 = 6.0\text{m/s}$, and the dynamic friction coefficient between the slider and the platform is $\mu = 0.25$. After reaching the platform's edge at point B, the slider flies horizontally out. Given that $AB = 2.2\text{m}$, the horizontal distance from the slider's landing point to the platform's edge is (). ($g = 10$)

m/s²)



- A. 3m
B. 1.5m
C. 1m
D. 2m

46 As shown in the figure, the parallel horizontal metal rails are connected to resistors R at both ends. The wire ab can slide frictionlessly on the frame, and a uniform magnetic field passes perpendicularly through the frame plane. When ab moves uniformly to the right, which of the following statements is correct?



A. Once the wire ab reaches a certain velocity, removing the force B . If the wire ab moves at a constant speed and the resistance R is increased, the wire ab can also maintain uniform motion on the frame. Therefore, the force pulling the wire ab can be doubled.

move

C. The higher the velocity of conductor ab , the lower the required pulling force. D. In addition to the pulling force, conductor ab is also subjected to magnetic field forces.

act on

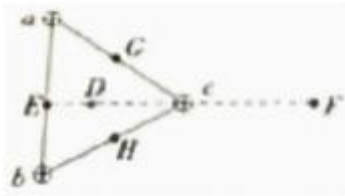
47. As shown in the figure, a uniform horizontal electric field is present on a horizontal tabletop. A small object with mass $m=0.1\text{kg}$ and charge $Q=-2.0\times 10^{-5}\text{C}$ is placed at point O on the tabletop and released from rest. Given the sliding friction force $f=0.2\text{N}$ between the object and the tabletop, the electric force $F=0.4\text{N}$ acts to the right. Calculate the work done by the resultant force on the object over 2 seconds after release.



- A. 3.2J
B. 0.8J
C. 2.4J
D. 1.6J

48 In a scientific experimental apparatus, scientists are investigating a unique charge distribution. They are working on a smooth absolute...

A triangular insulating pad model was placed on the horizontal plane, with three charge detectors positioned at vertices A, B, and C respectively. For further investigation, an electric field detector F was placed at point E, symmetrically about the zero-charge point. Which of the following statements is correct?



- A. The detector D reads zero B. The reading values of detector H vary
 C. The readings of detector D and detector F are the same. D. The detector D and F read the acceleration motion.

1 Which of the following physical quantities is **not** a vector?

- A. Magnetic induction intensity (vector, with magnitude and direction) B. Momentum (vector, $p=mv$, with magnitude and velocity direction)
C. Electric field strength (vector, with magnitude and direction) **D. Potential energy (scalar, only magnitude, no direction, such as gravity)**

potential energy, electric potential energy)

Analysis: A vector must possess both magnitude and direction.

2. The gravitational force acting on an object on Mars is one-third of its gravitational force on Earth. Therefore, the gravitational force on an object with mass $m=4.8\text{kg}$ on Mars is (). ($g=10\text{ m/s}^2$)

- A. 144N B. 24N
C. 48N **D. 16N** Analysis: Earth's gravity $G_{\text{Earth}} =$

$$mg = 4.8\text{kg} \times 10\text{m/s}^2 = 48\text{N}$$

$$\text{Mars gravity } G_{\text{Mars}} = \frac{1}{3} G_{\text{Earth}} = \frac{1}{3} \times 48\text{N} = 16\text{N}.$$

3. A spring with a spring constant $k=100\text{ N/m}$ is stretched from its original length of 10cm to 15cm under a tension force F. What is the magnitude of F?

- A. 5N** B. 25N
C. 10N D. 15N

Analysis: Hooke's Law states that $F = k\Delta x$, where:

The deformation of the beam is 5cm ($\Delta x = 15\text{cm} - 10\text{cm} = 0.05\text{m}$), and the spring constant is 100N/m.

The tension force F equals 100 multiplied by 0.05, resulting in 5N (the spring force balances the tension force, with equal magnitudes).

4. An object with mass $m=4\text{kg}$ is stationary on a horizontal surface, where the sliding friction coefficient between the surface and the object is $\mu=0.1$. When a horizontal force of 2N is applied to the object, the frictional force exerted on it is ().

- A. 2N** B. 3N
C. 4N D. 6N

Analysis: Maximum Static Friction Force of an Object

The maximum frictional force (f_{max}) equals $0.1 \times 4\text{kg} \times 10\text{m/s}^2 = 4\text{N}$. Since the horizontal tension is $2\text{N} < 4\text{N}$, the object remains stationary, and the static frictional force balances the tension, resulting in a static frictional force of 2N.

5. An object is subjected to three forces with magnitudes 1 = 2, 2 = 4, and 3 = 6. What is the maximum resultant force acting on the object?

- A. 12N** B. 0N
C. 10N D. 24N

Analysis: The maximum resultant force of three forces is when they are in the same direction, i.e., $2\text{N} + 4\text{N} + 6\text{N} = 12\text{N}$.

6 On a smooth horizontal surface, an object with mass $m=2\text{kg}$ is subjected to a horizontal force F, starting from rest and undergoing uniform acceleration. If its displacement within 2 seconds is 8m, then the magnitude of the horizontal force F is ().

- A. 10N B. 6N
C. 8N D. 2N

Analysis: From the formula for uniformly accelerated displacement $x = \frac{1}{2}at^2$, we get $a = \frac{2x}{t^2} = \frac{2 \times 8\text{m}}{(2\text{s})^2}$,

$$\text{Traction } F = ma = 2\text{kg} \times 4\text{m/s}^2 = 8\text{N}$$

7 An object moving in uniform acceleration on a smooth horizontal surface from rest has an acceleration of $a=2\text{m/s}^2$. What is its velocity at the end of 4 seconds?

- A. 4m/s B. 16m/s
C. 8m/s D. 2m/s Analysis: $v=at$, $v=2\text{m/s}^2 \times 4\text{s}=8\text{m/s}$

8 An object is dropped from a height of 100 meters. What is its height from the ground at 4 seconds? ($g=10 \text{ m/s}^2$)

- A.45m
C. 80m
B.10m
D.20m

Analysis: The height h of the fall within 4 seconds is $\frac{1}{2}gt^2 = \frac{1}{2} \times 10 \text{ m/s}^2 \times (4\text{s})^2 = 80\text{m}$, the height of the ground $100\text{m} - 80\text{m} = 20 \text{ m}$.

9 An object with mass $m=5\text{kg}$ is placed on a long smooth inclined plane with an inclination angle $\theta=30^\circ$. Starting from rest, the distance it travels in 2 seconds is (). ($g=10\text{m/s}^2$)

- A.5m
C.10m
B.1m
D.2m

Analysis: The acceleration $a = g\sin\theta = 10\text{m/s}^2 \times \sin 30^\circ = 5\text{m/s}^2$,
Displacement x in 2 seconds $= \frac{1}{2}at^2 = \frac{1}{2} \times 5\text{m/s}^2 \times (2\text{s})^2 = 10\text{m}$.

10. For a 2kg Object Subjected to Two Forces of 6N and 8N Respectively, What Is Its Maximum Acceleration?

- A.4m/s²
C.3m/s²
B.7m/s²
D.5m/s²

Analysis: The resultant force of the two forces is $6\text{N} + 8\text{N} = 14\text{N}$.

maximal acceleration $a = \frac{F}{m} = \frac{14\text{N}}{2\text{kg}} = 7\text{m/s}^2$

11. A 0.5-meter-long straight conductor is placed perpendicular to a uniform magnetic field. When a 10A current flows through it, the conductor experiences an Ampere force of 1.5N. Determine the magnetic induction intensity of the field.

- A.0.3T
C.0.1T
B.0.2T
D.0.4T

Analysis: Ampere force formula $F=BIL$

Get $B = \frac{F}{IL} = \frac{1.5\text{N}}{10\text{A} \times 0.5\text{m}} = 0.3\text{T}$,

12 A point moves in uniform circular motion with a velocity of $v=6 \text{ m/s}$ and a radius of $R=3 \text{ m}$. What is its angular velocity?

- A.2πrad/s
C. 2rad/s
B.3πrad/s
D.3rad/s

Analysis: Angular velocity $\omega = \frac{v}{R} = \frac{6\text{m/s}}{3\text{m}} = 2\text{rad/s}$.

13 As shown in the figure, an object moves from point A to point C via point B along two semicircles with a radius of $R=6\text{m}$ in a time of $t=4\text{s}$. What is the average velocity of the object?



- A.6m/s
C.6πm/s
B.12m/s
D.3πm/s

Calculation: Average velocity = displacement divided by time. Given the displacement is $2R = 12 \text{ meters}$, the velocity is calculated as $v = \frac{12\text{m}}{4\text{s}} = 3\text{m/s}$.

14. Which of the following statements about distance and displacement is correct?

- A. Distance is a vector, displacement is a scalar.
- B. The magnitude of displacement is not necessarily equal to the distance traveled.
- C. Both distance and displacement are vectors

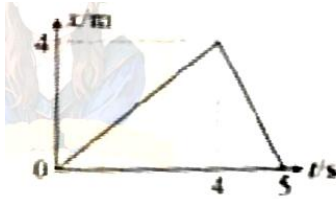
D. The magnitude of displacement can be equal to the distance traveled.

15. An object with mass $m=0.2\text{kg}$ moves at constant velocity $v=20\text{m/s}$ on a smooth horizontal surface. What is its momentum?

- A. $0.8\text{kg} \cdot \text{m/s}$
- B. $4\text{kg} \cdot \text{m/s}$
- C. $0.4\text{kg} \cdot \text{m/s}$
- D. $8\text{kg} \cdot \text{m/s}$

Analysis: The momentum $p=mv=0.2\text{kg} \times 20\text{ m/s} = 4\text{ kg}\cdot\text{m/s}$

16. The graph of the displacement (x) of a toy car moving in a straight line versus time (t) is shown in the figure. Which of the following statements is correct?



- A. Within 0 to 4 seconds, the toy car moves a distance of 8 meters.
- B. Between 4 and 5 seconds, the toy car moves at a speed of 0.8 meters per second.
- C. The toy car's speed is 1.0m/s in $0\sim 4\text{s}$
- D. The toy car's average speed is 0.8m/s in $0\sim 5\text{s}$**

17. An object with mass $m=5\text{kg}$ is placed on a smooth inclined plane with an angle of inclination $\theta=30^\circ$ and a height of 20m . When it starts sliding from rest, what is the magnitude of the gravitational impulse it experiences within 2 seconds? ($g=10\text{m/s}^2$)

- A. $50\text{N} \cdot \text{s}$
- B. $25\text{N} \cdot \text{s}$
- C. $200\text{N} \cdot \text{s}$
- D. $100\text{N} \cdot \text{s}$**

Analysis: The gravitational impulse $I=mgt=5\text{kg}\times 10\text{m/s}^2 \times 2\text{s}=100\text{N}\cdot\text{s}$

18. When a 10kg object is pulled horizontally by a force F of 20N , it moves 5m on a rough horizontal surface with a frictional force $f=5\text{N}$. The work done by the pulling force F is ().

- A. 100J**
- B. 75J
- C. -25J
- D. 25J

Analysis: The work done by the pulling force is $W=Fs=20\text{N}\times 5\text{m}=100\text{J}$.

19. A 3kg ball on a smooth surface is subjected to an external force that increases its velocity from 2 m/s to 6 m/s . Calculate the work done by this force.

- A. 54J
- B. 60J
- C. 48J**
- D. 6J

Analysis: The work-energy theorem $W=\Delta E_k=\frac{1}{2}mv_2^2-\frac{1}{2}mv_1^2=\frac{1}{2}\times 3\times 6^2-\frac{1}{2}\times 3\times 2^2=48\text{J}$

20. When light enters a medium from air at an incident angle $i=60^\circ$, the refractive angle $r=30^\circ$, and the refractive index of air is $n=1$, the refractive index of the medium is ().

- A. 2
- B. $\sqrt{3}$**
- C. 5
- D. 1.5

Analysis: Refraction law $n_1 \sin i = n_2 \sin r$, get $n_2 = \frac{\sin i}{\sin r} = \frac{\sin 60^\circ}{\sin 30^\circ} = 2$

21. A spring oscillator undergoing simple harmonic motion completes one full cycle in 2 seconds. What is its frequency?

- A. 2 Hz
C. 0.2 Hz
B. 0.5 Hz
D. 5 Hz

Analysis: Frequency $f = \frac{1}{T} = \frac{1}{2} = 0.5 \text{ Hz}$.

22. A sound wave with a frequency of $f = 200 \text{ Hz}$ travels through air at a speed of $v = 340 \text{ m/s}$. What is the wavelength of this sound wave?

- A. 1.7 m
C. 6.8 m
B. 3.4 m
D. 680 m

Analysis: The wavelength $\lambda = \frac{v}{f} = \frac{340}{200} = 1.7 \text{ m}$.

23. Each question has 4 options, and only 1 is correct.

In a vacuum, two point charges are separated by a distance d , with a Coulomb force of F . If the distance between the charges is kept constant and the charge of one of them is doubled, the Coulomb force between them becomes ().

- A. 2 times
C. $\frac{1}{4}$ times
B. 4 times
D. $\frac{1}{2}$ times

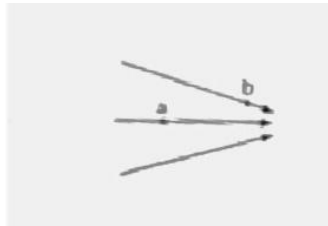
Analysis: Coulomb's Law $F = k \frac{q_1 q_2}{d^2}$. When the charge doubles, the force doubles.

24. The velocity of a water drop with zero initial velocity falling from a height of 5 m is ($g = 10 \text{ m/s}^2$) ()

- A. 10 m/s
C. 2 m/s
B. 5 m/s
D. 20 m/s

Analysis: From $v^2 = 2gh$, we get $v = \sqrt{2 \times 10 \times 5} = 10 \text{ m/s}$.

26. The electric field lines are distributed as shown in the figure. If equal point charges are placed at points a and b within the electric field, the relationship between the electric forces acting on these charges is ().



- A. $\geq$
C. $<$
B. =
D. $>$

The denser the electric field lines, the stronger the field strength, with $E_a > E_b$. According to the formula $F = qE$, this implies $F_a > F_b$.

The total resistance of two resistors, 20Ω and 30Ω , in parallel is ().

- A. 50Ω
C. 10Ω
B. 60Ω
D. 12Ω

Analysis: The total parallel resistance R is $\frac{R_1 R_2}{R_1 + R_2} = \frac{20 \times 30}{20 + 30} = 12 \Omega$.

27. Two resistors with resistance values of $R_1 = 40 \Omega$ and $R_2 = 30 \Omega$ are connected in parallel and connected to a circuit with a voltage of 12 V. The current ratio $I_1 : I_2$ through the two resistors R_1 and R_2 is ()

- A. 1:1
C. 4:3
B. 1:2
D. 3:4

Analysis: The voltage of the parallel circuit is equal, $I_1 : I_2 = R_2 : R_1 = 30 \Omega : 40 \Omega = 3:4$.

28. For an ideal gas of a given mass, if the temperature remains constant and the pressure decreases, its volume will ().

- A. invariant
B. diminution
C. All of the above answers are possible.
D. largen

Analysis: Boyle's law $pV=C$ (constant temperature), the pressure is small, the volume is large.

29. A point charge of $3.0 \times 10^{-9} \text{ C}$ is placed in an electric field of $6.0 \times 10^4 \text{ N/C}$. The magnitude of the electric force acting on the point charge is ().

- A. $1.8 \times 10^{-4} \text{ N}$
B. $2.0 \times 10^{-4} \text{ N}$
C. $1.8 \times 10^4 \text{ N}$
D. $1.8 \times 10^{-5} \text{ N}$

Analysis: The electric force $F=qE=3.0 \times 10^{-9} \times 6.0 \times 10^4 = 1.8 \times 10^{-4} \text{ N}$

30. The international units of magnetic induction B and magnetic flux are ().

- A. Tesla, Weber
B. Amperes, volts
C. Pascal, Hertz
D. Joules, watts

31. When a beam of light strikes the air-glass interface at an angle of 30° , the angle between the reflected and incident rays is ().

- A. 30°
B. 45°
C. 60°
D. 80°

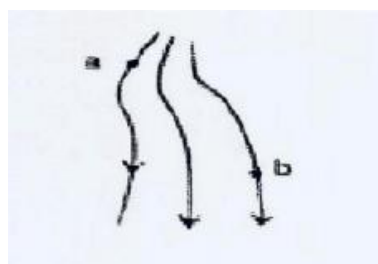
Analysis: The reflection angle equals the incident angle, both being 30° . The angle between them is $30^\circ + 30^\circ = 60^\circ$.

32. A 0.2-meter-long wire carrying a constant current is placed in a uniform magnetic field with a magnetic induction intensity of 0.8 T. If the maximum ampere force acting on the wire in the magnetic field is 0.08 N, the current flowing through the wire is ().

- A. 0.5A
B. 5A
C. 0.2A
D. 2A

Analysis: The maximum Ampere force F equals the product of the magnetic field (B) and the length of the wire (L), with the wire perpendicular to the magnetic field. $I = \frac{F}{BL} = \frac{0.08}{0.8 \times 0.2} = 0.5 \text{ A}$.

33. An ideal gas of a certain mass changes from state A to state B. The process $A \rightarrow B$ is an isobaric process (constant pressure). Given that the volume of state A is $V_1 = 1.5 \text{ m}^3$, the temperature of state A is $T_1 = 300 \text{ K}$, and the temperature of state B is $T_2 = 400 \text{ K}$, the volume of state B is ()



- A. 1.5 m^3
B. 4.5 m^3
C. 2 m^3
D. 6 m^3

Analysis: Gay-Lussac Law $\frac{V_1}{T_1} = \frac{V_2}{T_2}$, get $V_2 = \frac{V_1 T_2}{T_1} = \frac{1.5 \times 400}{300} = 2 \text{ m}^3$.

35. A ball of mass $m_1 = 20 \text{ kg}$ is moving at $v = 6 \text{ m/s}$ and collides with a car that is stationary on a smooth horizontal surface. The car has a mass of 40 kg. What is the combined velocity of the ball and the car after the collision?

- A. 2m/s
B. 3m/s
C. 4m/s
D. 5m/s

Analysis: Conservation of momentum $m_1 v = (m_1 + m_2) v$

get $v_{\text{共}} = \frac{20 \times 6}{20 + 40} = 2 \text{ m/s}$

36. A basketball with a mass of 0.5 kg is launched from the ground to a height of 1 meter, then falls back to the ground. The change in gravitational potential energy during this entire process is ().

- A. 5J
B. 0.5J
C. 10J
D. 0J

Analysis: The change in gravitational potential energy equals the final potential energy minus the initial potential energy. Since both initial and final positions are at ground level, the change is 0J.

37. As shown in the figure, a thin rope is fixed at point O at one end and suspended from the other end a ball A with a mass of $m=6\text{kg}$. When the rope forms a 30-degree angle with the vertical direction and the ball A is at rest, the horizontal force applied to the ball A should be () ($g=10\text{m/s}^2$).

- A. 10 N
B. 12N
C. $20\sqrt{3}\text{N}$
D. 20N

Analysis: The horizontal tension $F = mg \tan 30^\circ = 6 \times 10 \times \frac{1}{\sqrt{3}} = 20\sqrt{3}\text{N}$ is balanced.

38. The diagram shows two charged metal spheres A and B in a vacuum, each carrying a charge of $Q = \pm 3.0 \times 10^{-4}\text{C}$, with a distance of $r=6\text{cm}$ between them. Given the electrostatic constant $K = 9 \times 10^9 \text{N}\cdot\text{m}^2/\text{C}^2$, the electric field strength at the midpoint between the spheres is

- A. $1.5 \times 10^4 \text{N/C}$
B. 0N/C
C. $6.0 \times 10^4 \text{N/C}$
D. $3.0 \times 10^4 \text{N/C}$

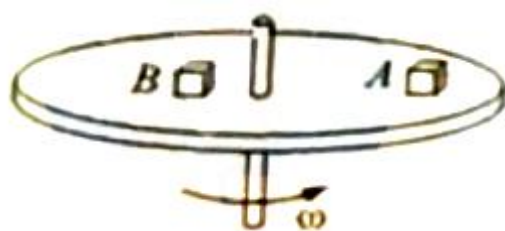
Analysis: A and B are different charges. The field strength at the midpoint is the superposition of the field strengths of the two charges, $E = K \frac{Q}{r^2} + k \frac{Q}{r^2} = 4k \frac{Q}{r^2} = 6.0 \times 10^4 \text{N/C}$

39. A small ball is thrown horizontally from a building 20m high at a speed of $v_0 = 20\text{m/s}$. The air resistance is not considered. Find the velocity of the ball when it lands. ($g = 10\text{m/s}^2$)

- A. 40 m/s
B. $20\sqrt{2}\text{m/s}$
C. 40m/s
D. 20m/s

Analysis: Vertical velocity $v_y = 2gh = 2 \times 10 \times 20 = 20\text{m/s}$, resultant velocity $v = \sqrt{v_0^2 + v_y^2} = 20\sqrt{2}\text{m/s}$

40. As shown in the figure, two objects A and B with equal mass are placed on a horizontal disk rotating uniformly around a vertical axis. If the two objects remain stationary relative to the disk, then the two objects ().



A. The angular velocity is the same.

The angular velocity of B is greater than that of A.

C. The angular velocity of A is less than that of B. D. The linear velocity is the same.

41. A point charge with a charge of $+4 \times 10^{-6}\text{C}$ is moved from point A to point B in the electric field. The work done by the electric field force is $-8 \times 10^{-4}\text{J}$. The electric potential difference between AB is ()

$$U = \frac{W}{q} = \frac{-8 \times 10^{-4}}{4 \times 10^{-6}} = -200\text{V}$$

A. -200V
C. -2000V

B. 200V
D. 2000V

Analysis: The potential difference $U_{AB} = \frac{W_{AB}}{q} = \frac{-8 \times 10^{-4}}{-6} = -200V$

$q = 4 \times 10^{-6}C$

42. Consider a point charge conductor A carrying $+8q$. At a distance r from A, there is another identical conductor B carrying $-4q$. When A and B are brought into contact and then returned to their original positions, the Coulomb force between them will be restored to its original value.

- A. $\frac{1}{4}$ times
B. 4 times
C. 8 times

D. $\frac{1}{8}$ times

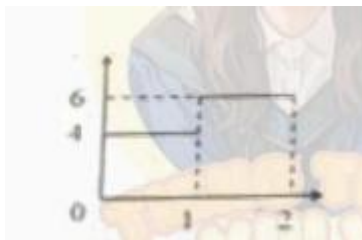
Analysis: After contact, the charges are neutralized and then divided equally, $q_A = q_B = \frac{8q - 4q}{2} = 2q$. The original Coulomb force $F_1 = k \frac{8q \times 4q}{r^2} = 32k \frac{q^2}{r^2}$,

Later $F_2 = k \frac{2q \times 2q}{r^2} = 4k \frac{q^2}{r^2}$, $F_2 = \frac{F_1}{8}$. 43. A bullet traveling at 700 m/s horizontally penetrates the first fixed wooden board, reducing its velocity to 500 m/s. If it continues through the second identical wooden board, its velocity becomes ().

- A. 200 m/s
B. 100 m/s
C. 50 m/s
D. 300 m/s

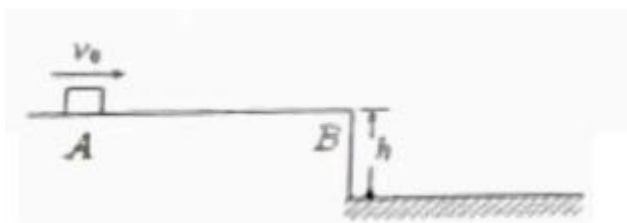
Analysis: The work done by a body passing through two wooden boards is the same according to the work-energy theorem., $\frac{1}{2}mv_1^2 - \frac{1}{2}mv_2^2 = \frac{1}{2}mv_2^2 - \frac{1}{2}mv_3^2$, Substituting, we get $v_3 = 100$ m/s

44. A 2kg object starts from rest on a smooth horizontal surface and is subjected to a horizontal force. The graph of the force versus time is shown in the figure. What is the object's kinetic energy at the end of 2 seconds? (A) 25 (B) 16 (C) 36 (D) 50



- A. 25
B. 16
C. 36
D. 50

45. As shown in the figure, the horizontal platform AB is $h=0.80$ m above the ground. A slider moves in uniform acceleration along the platform from point A with an initial velocity of $v_0=6.0$ m/s, and the dynamic friction coefficient between the slider and the platform is $\mu=0.25$. After reaching the platform's edge at point B, the slider flies horizontally out. Given that $AB=2.2$ m, the horizontal distance from the slider's landing point to the platform's edge is (). ($g=10$ m/s²)

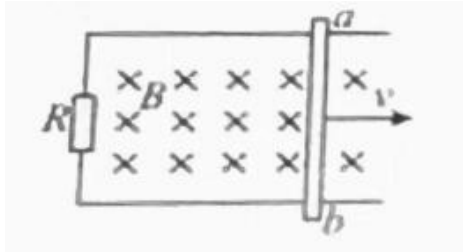


- A. 3m
B. 1.5m
C. 1m
D. 2m acceleration: $a = -\mu g = -2.5 \text{ m/s}^2$
B point velocity: $v_B^2 = 6^2 + 2 \times (-2.5) \times 2.2 = 25, v_B = 5 \text{ m/s}$

time of flat throw: $t = \sqrt{\frac{2h}{g}} = \sqrt{\frac{2 \times 0.8}{10}} = 0.4\text{s}$

Horizontal distance: $x = 5 \times 0.4 = 2\text{m}$

46 As shown in the figure, the parallel horizontal metal rails are connected to resistors R at both ends. The wire ab can slide frictionlessly on the frame, and a uniform magnetic field passes perpendicularly through the frame plane. When ab moves uniformly to the right, which of the following statements is correct?



A. Once the wire ab reaches a certain velocity, removing the force B . If the wire ab moves at a constant speed and the resistance R is increased, the wire ab can also maintain uniform motion on the frame. Therefore, the force pulling the wire ab can be doubled.

move

C. The higher the velocity of conductor ab , the less the required pulling force **D. In addition to the pulling force, conductor ab is also subjected to magnetic field force.**

act on

47. As shown in the figure, a uniform electric field exists horizontally on a tabletop. A small object with mass $m=0.1\text{kg}$ and charge $Q=-2.0 \times 10^{-5}\text{C}$ is placed at point O on the tabletop and released from rest. Given the sliding friction force $f=0.2\text{N}$ between the object and the tabletop, the electric force $F=0.4\text{N}$ acts to the right. Calculate the work done by the resultant force on the object over 2 seconds after release.



A.3.2J

B.0.8J

C.2.4J D.1.6J **Combined external force $F_{combined} =$**

$0.4-0.2=0.2\text{N}$

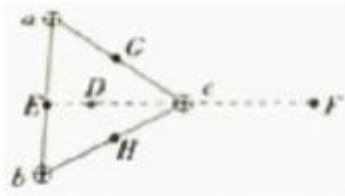
accelerated speed $a = \frac{0.2}{0.1} = 2\text{m/s}^2$

gression $x = \frac{1}{2} \times 2 \times 2^2$

The work done by the external force is $W_0 = 0.3 \times 4 = 0.8\text{J}$.

48 In a scientific experimental apparatus, scientists are investigating a unique charge distribution. They are working on a smooth absolute...

A triangular insulating pad model was placed on the horizontal plane, with three charge detectors positioned at vertices A, B, and C respectively. For further investigation, an electric field detector F was placed at point E, symmetrically about the zero-charge point. Which of the following statements is correct?



A. The detector D reads zero

B. The reading values of detector H vary

C. The readings of detector D and detector F are the same.

D. The detector D and F read the acceleration motion.